

Grid Combi and other fun problems

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1 Usual Advice

- Problem difficulty in Combi is very relative, a 35 MOHS could easily feel like a 5 or 10 MOHS.
- You really don't need advanced techniques for Combi (except for maybe graph theory). Just a deep understanding of the problem and its dynamics.
- Core concepts overlap in many situations: what you see as a graph theory problem might also be seen as a double counting problem and so on...
- When a problem asks for a minimal or maximal value s , you provide a working configuration for $p = s$ then prove $p \geq s$. One of the two parts is usually much harder than the other, make sure you work on them both.
- Consider extremal (Maxima and Minima) arguments. The Convex Hull is useful when you have a set of points on the plane for example.
- Mean value theorem can be used when discussing paths on the plane.
- Coloring is very powerful when dealing with graphs or grids.
- Always try to look for symmetry.
- Always pause to consider your options! Don't chase induction for one hour straight, it might be a dead end!

2 Group Solving

Trust me it's easier than you expect.

- **ISL23/C6** Let N be a positive integer, and consider an $N \times N$ grid. A right-down path is a sequence of grid cells such that each cell is either one cell to the right of or one cell below the previous cell in the sequence. A right-up path is a sequence of grid cells such that each cell is either one cell to the right of or one cell above the previous cell in the sequence.

Prove that the cells of the $N \times N$ grid cannot be partitioned into less than N right-down or right-up paths.

A hint will be given every 15 minutes.

3 Problems

You should be able to solve all of these. All are fun and have nice solutions, you can enjoy them on a vehicle trip or while going on a walk :)

- **ISL17/C1** A rectangle $\mathcal{R}$ with odd integer side lengths is divided into small rectangles with integer side lengths. Prove that there is at least one among the small rectangles whose distances from the four sides of $\mathcal{R}$ are either all odd or all even.

- **USAMO00/4** Find the smallest positive integer n such that if n squares of a 1000×1000 chessboard are colored, then there will exist three colored squares whose centers form a right triangle with sides parallel to the edges of the board.
- **USAMO05/5** Let n be an integer greater than 1. Suppose $2n$ points are given in the plane, no three of which are collinear. Suppose n of the given $2n$ points are colored blue and the other n colored red. A line in the plane is called a balancing line if it passes through one blue and one red point and, for each side of the line, the number of blue points on that side is equal to the number of red points on the same side. Prove that there exist at least two balancing lines.
- **India04** The game of *pebbles* is played on an infinite board of lattice points (i, j) . Initially there is a *pebble* at $(0, 0)$. A move consists of removing a *pebble* from point (i, j) and placing a *pebble* at each of the points $(i + 1, j)$ and $(i, j + 1)$ provided both are vacant. Show that at any stage of the game there is a *pebble* at some lattice point (a, b) with $0 \leq a + b \leq 3$.
- **TSTST16/5** In the coordinate plane are finitely many walls; which are disjoint line segments, none of which are parallel to either axis. A bulldozer starts at an arbitrary point and moves in the $+x$ direction. Every time it hits a wall, it turns at a right angle to its path, away from the wall, and continues moving. (Thus the bulldozer always moves parallel to the axes.)
Prove that it is impossible for the bulldozer to hit both sides of every wall.
- **IMO23/5** Let n be a positive integer. A Japanese triangle consists of $1 + 2 + \cdots + n$ circles arranged in an equilateral triangular shape such that for each $i = 1, 2, \dots, n$, the i^{th} row contains exactly i circles, exactly one of which is coloured red. A ninja path in a Japanese triangle is a sequence of n circles obtained by starting in the top row, then repeatedly going from a circle to one of the two circles immediately below it and finishing in the bottom row. Here is an example of a Japanese triangle with $n = 6$, along with a ninja path in that triangle containing two red circles. In terms of n , find the greatest k such that in each Japanese triangle there is a ninja path containing at least k red circles.
- **ToT F99 J-A6(Evan Chen phrasing)** A rook is allowed to move one cell either horizontally or vertically. After 64 moves the rook visited all cells of the 8×8 chessboard and returned back to the initial cell. Prove that the number of moves in the vertical direction and the number of moves in the horizontal direction cannot be equal.
- **Bonus - IMO24/5** Turbo the snail plays a game on a board with 2024 rows and 2023 columns. There are hidden monsters in 2022 of the cells. Initially, Turbo does not know where any of the monsters are, but he knows that there is exactly one monster in each row except the first row and the last row, and that each column contains at most one monster.
Turbo makes a series of attempts to go from the first row to the last row. On each attempt, he chooses to start on any cell in the first row, then repeatedly moves to an adjacent cell sharing a common side. (He is allowed to return to a previously visited cell.) If he reaches a cell with a monster, his attempt ends and he is transported back to the first row to start a new attempt. The monsters do not move, and Turbo remembers whether or not each cell he has visited contains a monster. If he reaches any cell in the last row, his attempt ends and the game is over.
Determine the minimum value of n for which Turbo has a strategy that guarantees reaching the last row on the n -th attempt or earlier, regardless of the locations of the monsters.
- **Bonus - TSTST19/3** On an infinite square grid we place finitely many cars, which each occupy a single cell and face in one of the four cardinal directions. Cars may never occupy the same cell. It is given that the cell immediately in front of each car is empty, and moreover no two cars face towards each other (no right-facing car is to the left of a left-facing car within a row, etc.). In a move, one chooses a car and shifts it one cell forward to a vacant cell. Prove that there exists an infinite sequence of valid moves using each car infinitely many times.

4 Hints

- 7We draw a circular path on the edge squares, then another one at the level under that etc... How can this drawing with 4 paths be transformed into one connected path?
- 5Consider the points on the 'convex hull' of all mirrors and leverage the fact that the bulldozer only goes infinitely in one direction.
- 2Set a condition and check what would happen at the last columns.
- 6Draw the tree of possible paths. It's just pigeonhole.
- 1Coloring
- 3Consider the convex hull
- 4Look for an invariant

Purposfully didn't provide hints for bonus problems